

Project Title: Blockchain Land Registry System

Course Name: Foundations of Blockchain Technology

Course Code: BCSE324L

Faculty Name: Dr. Deepika Roselind J

Students Name and Registration no:

Achal Pramod Tripathi (23BCE1734)

Mayank (23BCE1753)

Abstract

Traditional land registry systems are often centralized, opaque, and reliant on paper-based processes, making them vulnerable to fraud, data loss, and inefficiency. This project presents a Blockchain-Based Land Registry System, a decentralized application (DApp) designed to address these challenges. By leveraging the inherent security and transparency of blockchain technology, this system creates a single, immutable source of truth for property ownership. The DApp is built on the Ethereum blockchain, with a smart contract written in Solidity to manage all core logic. Key features include secure property registration, where document proof is stored immutably on the InterPlanetary File System (IPFS), an administrative verification layer for legal validation, a secure two-step ownership transfer process, and a novel Mortgage & Lien System. The user interface is a modern React web application that interacts with the blockchain via a user's MetaMask wallet, ensuring that all actions are cryptographically signed and authorized. This project demonstrates a practical and superior model for managing real-world assets, offering enhanced security, transparency, and efficiency over conventional methods.

Overview of the project

Introduction

The management of land and property records is a cornerstone of a stable economy. However, traditional systems are plagued by issues such as bureaucratic bottlenecks, forged documents, and a lack of public transparency, leading to disputes and significant financial losses. The advent of blockchain technology offers a revolutionary alternative. By distributing a digital ledger across a network of computers, no single entity can alter or destroy records, and all transactions are publicly verifiable.

Objective

The primary objective of this project is to design, develop, and deploy a fully functional, end-to-end decentralized application for land and property registration. The system aims to replace the ambiguity and insecurity of traditional registries with a transparent, immutable, and cryptographically secure alternative.

Scope & Core Features

The project is built as a complete DApp with three main layers: a frontend, a backend API, and a smart contract. The system's scope is defined by the following five major features:

1. **Property Registration:** Authenticated users (via MetaMask) can register a new property by providing its location, area, and official documents (e.g., deeds, surveys).
2. **Document Storage (IPFS):** To ensure data immutability and keep blockchain storage costs low, property documents are not stored on-chain. Instead, they are uploaded to the InterPlanetary File System (IPFS), and only the unique, permanent content hash (CID) is stored in the smart contract.
3. **Admin Verification:** A special "Admin" role (the contract owner) mimics a government land registry office. This admin has the exclusive authority to review and "verify" a property, marking it as legally valid on the blockchain.
4. **Secure Ownership Transfer:** The system implements a secure, two-step process for transferring property. The current owner must first *initiate* the transfer to a specific buyer's wallet address. The buyer must then *confirm* the transfer, preventing accidental or unauthorized changes in ownership.

5. **Mortgage & Lien System:** A sophisticated, real-world feature where a property owner can place a "lien" on their verified property as collateral for a loan (mortgage). This lien is publicly recorded, and the smart contract **blocks any sale or transfer** of the property until the lender (the bank or individual) signs a transaction to "clear the lien."

Methodology

The project was developed using a modern technology stack chosen to optimize for security, decentralization, and user experience.

Technology Stack

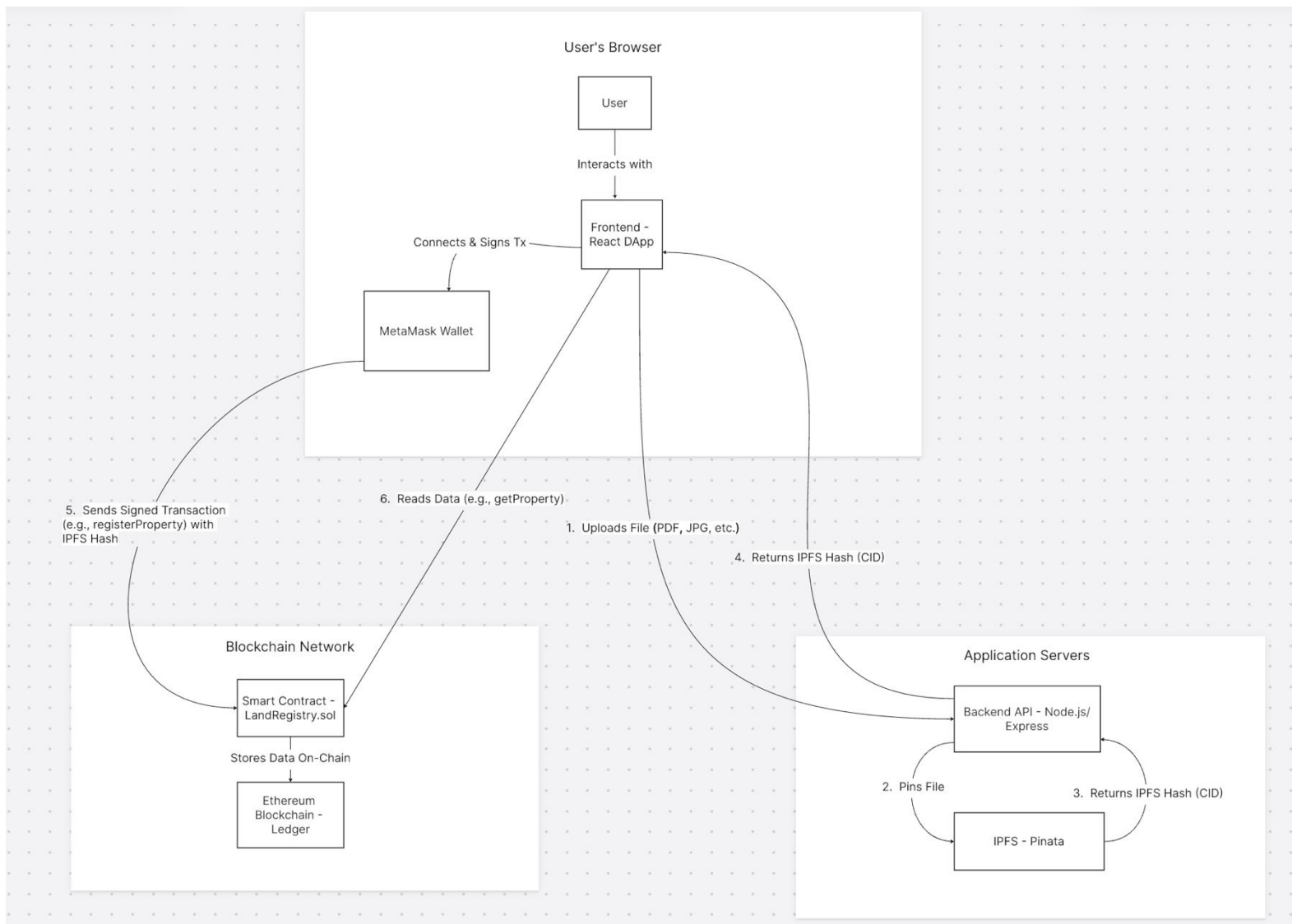
- **Blockchain & Smart Contract:**
 - **Solidity:** The primary language used to write the LandRegistry.sol smart contract, which contains all the business logic for properties, transfers, and liens.
 - **Ethereum (Sepolia Testnet):** The chosen public test network for deploying and testing the DApp in a realistic, decentralized environment.
 - **OpenZeppelin Contracts:** Utilized the industry-standard Ownable contract to securely manage the "Admin" role.
- **Development Environment:**
 - **Hardhat:** The core development suite used for compiling, deploying, running local nodes, and testing the smart contract.
 - **Chai:** The assertion library used in LandRegistry.test.js to write comprehensive unit tests covering all functions and security modifiers.
- **Frontend (User Interface):**
 - **React (with TypeScript):** A modern JavaScript library used to build a fast, responsive, and type-safe single-page application (SPA).
 - **Ethers.js:** A complete JavaScript library for interacting with the Ethereum blockchain. It is used to connect to the user's wallet, read data from the contract, and send transactions (e.g., registerProperty, placeLien).

- **MetaMask:** The primary method for user authentication. Users connect their personal MetaMask wallet, which acts as their digital identity and is used to sign all state-changing transactions.
- **Backend & File Storage:**
 - **Node.js (Express):** A lightweight backend server was created for a single, critical purpose: to securely handle document uploads. It acts as a bridge between the frontend and IPFS.
 - **IPFS (via Pinata):** The chosen decentralized storage solution. The backend API uses the Pinata service to pin files to IPFS, guaranteeing their persistence and returning the CID to be stored on the blockchain. This separation prevents secure API keys from being exposed in the frontend code.

Development Workflow

1. **Smart Contract Design:** The LandRegistry.sol contract was designed first, defining the core data structs (Property, Lien) and the functions (registerProperty, initiateTransfer, placeLien, etc.).
2. **Test-Driven Development (TDD):** Unit tests were written in parallel using Hardhat and Chai to validate every require statement, event emission, and state change.
3. **Deployment:** The contract was compiled and deployed to the Sepolia testnet using a Hardhat script (deploy.js). The resulting contract address was then configured in the environment files.
4. **Frontend & UI:** The React application was built component-by-component (e.g., RegisterProperty.tsx, MortgageManager.tsx), with Ethers.js handling all blockchain communication.
5. **Backend API:** A simple Node.js/Express endpoint (/upload-doc) was created to receive file data from the frontend and securely manage the upload to Pinata.
6. **Integration & Testing:** All three layers were integrated and tested end-to-end to ensure a seamless flow, from a user uploading a document to it being successfully registered on-chain.

Architecture Diagram



Detailed results (Output screenshots)

The project's success is confirmed by the following functional outputs, as documented in the provided screenshots.

Exhibit A: Welcome & Wallet Connection

- **Result:** The application presents a clean welcome screen, prompting the user to connect their wallet.

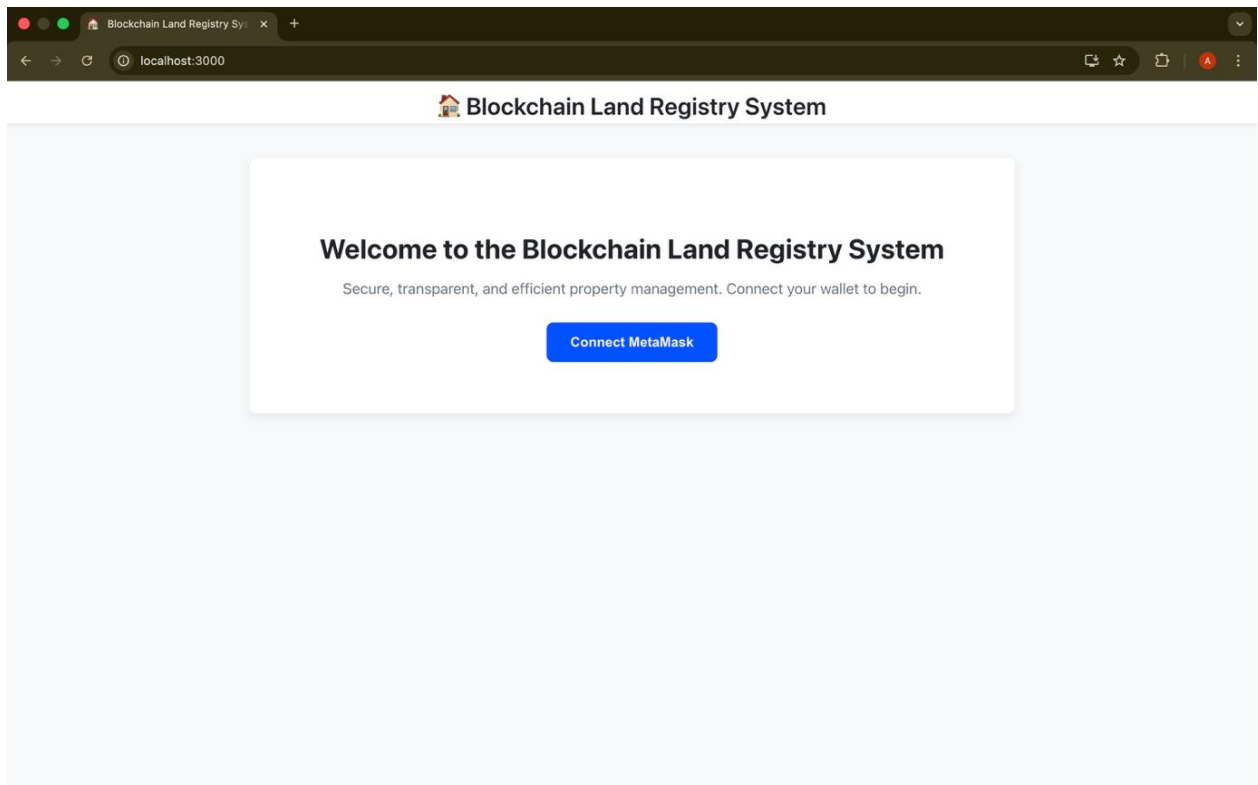


Exhibit B: Main User Dashboard

- **Result:** Upon successful connection, the user is presented with the main application dashboard, showing their wallet address, Admin status (if applicable), and all available features.

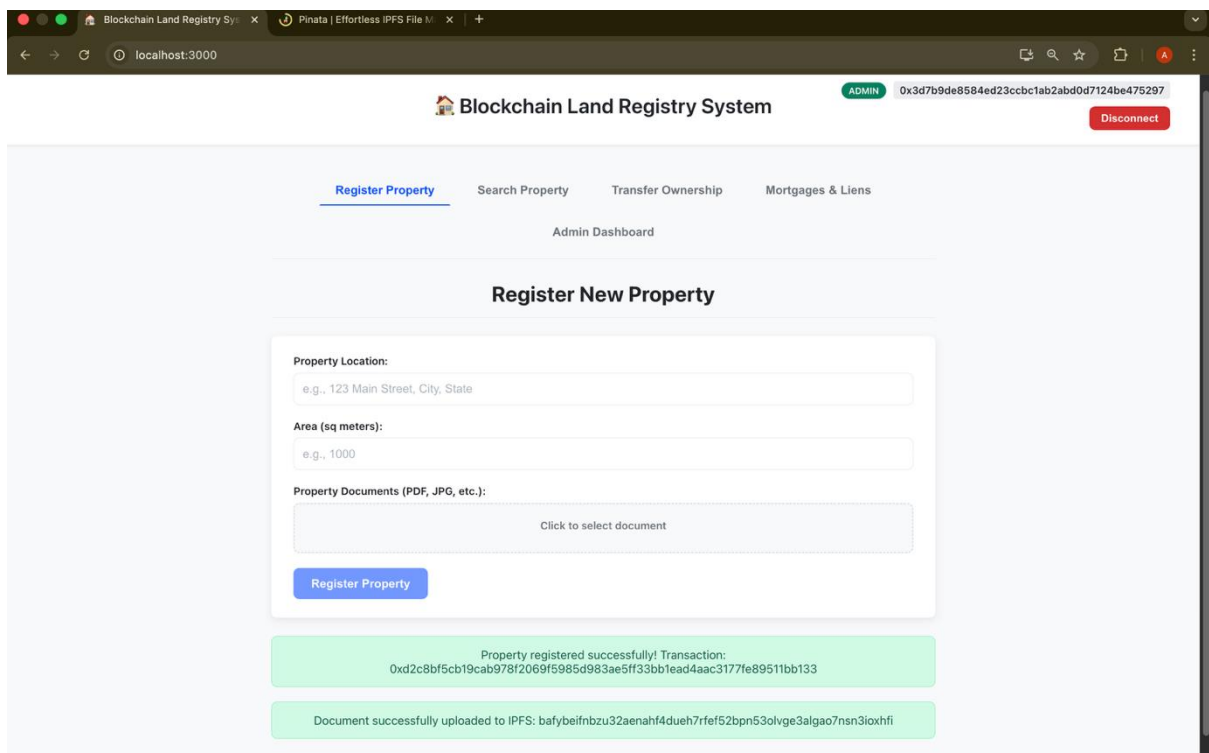
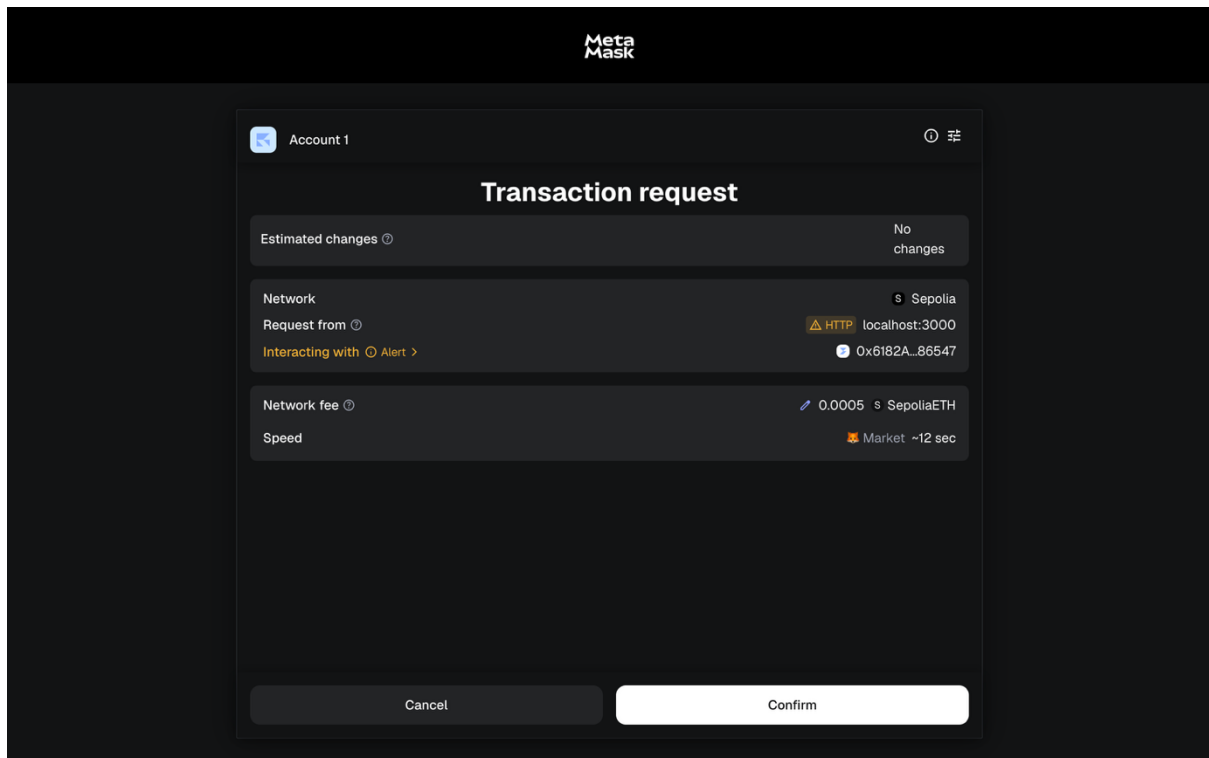
The screenshot displays a web browser window with the address bar showing 'localhost:3000'. The page title is 'Blockchain Land Registry System'. In the top right corner, there is a green 'ADMIN' button, a wallet address '0x3d7b9de8584ed23ccbc1ab2abd0d7124be475297', and a red 'Disconnect' button. The main navigation bar includes links for 'Register Property' (which is underlined), 'Search Property', 'Transfer Ownership', and 'Mortgages & Liens'. Below the navigation bar, the text 'Admin Dashboard' is centered. The main content area features a 'Register New Property' form. This form contains three input fields: 'Property Location:' with a placeholder 'e.g., 123 Main Street, City, State', 'Area (sq meters):' with a placeholder 'e.g., 1000', and 'Property Documents (PDF, JPG, etc.):' with a dashed border and the text 'Click to select document'. At the bottom of the form is a blue 'Register Property' button.

Exhibit C: Property Registration

- **Result:** The "Register Property" tab provides a clear form for submitting location, area, and documents, forming the entry point for all assets.

The screenshot shows a web browser window with the address bar displaying 'localhost:3000'. The page title is 'Blockchain Land Registry System'. The user is logged in as 'ADMIN' with a wallet address '0x3d7b9de8584ed23ccbc1ab2abd0d7124be475297'. A 'Disconnect' button is visible in the top right corner. The main navigation bar includes 'Register Property' (active), 'Search Property', 'Transfer Ownership', and 'Mortgages & Liens'. Below the navigation bar is an 'Admin Dashboard' section. The central content area is titled 'Register New Property' and contains a form with the following fields: 'Property Location' (filled with 'Room 919, A Block, VIT Chennai'), 'Area (sq meters)' (filled with '1500'), and 'Property Documents (PDF, JPG, etc.):' (filled with 'Software Engineering 2.png'). A blue 'Register Property' button is at the bottom of the form.

This screenshot shows the same 'Register New Property' form as the previous one, but with a 'Processing...' button instead of 'Register Property'. Below the form, there are two green status messages: 'Step 2/2: Registering property on the blockchain...' and 'Document successfully uploaded to IPFS: bafybeifnbzu32aenahf4dueh7rfe52bpn53olvge3algao7nsn3ioxhfi'. The rest of the interface, including the browser window and navigation bar, remains the same.



Blockchain Land Registry Sy...Pinata | Effortless IPFS File M...+

app.pinata.cloud/ipfs/files

Pinata

Workspace

Achal's Workspace

IPFS

Files

Groups

Gateways

Analytics

DEVELOPER

API Keys

Webhooks

Access Controls

EXTENSIONS

Marketplace

Integrations

FILES

PUBLIC

PRIVATE

Public files are accessible via IPFS

Search files and CIDs

	NAME	CID	SIZE	CREATION DATE	FILE ID
☐	Software Engineering 2.png	bafyb...oxhfi	339.34 KB	11/6/2025	
☐	Screenshot 2025-11-01 at 3.05.24 PM.png	bafkr...dq75e	91.63 KB	11/1/2025	
☐	GenAI Hackathon V1.1(Problem Statement).docx	bafkr...erenq	30.06 KB	11/1/2025	
☐	Sem 5 TT.pdf	bafkr...wb4c4	156.04 KB	11/1/2025	
☐	16. Winter Semester 2_ndaar - Except Law.pdf	bafyb...b24am	445.29 KB	11/1/2025	
☐	Ex-13 Softwae Engineering (1).pdf	bafyb...ovhce	790.79 KB	9/25/2025	
☐	680985832-wildlife-ec_ptel-quiz-answers.pdf	bafyb...xljci	396.33 KB	9/24/2025	
☐	Achal Tripathi Resume.pdf	bafyb...hy4rm	314.25 KB	9/23/2025	

Rows per page: 10

Documentation

Blockchain Land Registry Sy...Pinata | Effortless IPFS File M...+bafybeifnbzu32aenahf4dueh7ref52bpn53olvge3algao7nsn3ioxhfi

Lab Only						
L57+L58	AB1-514	SCOPE14	Clashed with TA2	60	2	58
L3+L4	AB1-514	SCOPE6	Similar to Registered Slot	60	4	56
L15+L16	AB1-514	SCOPE13	NONE	60	7	53
L19+L20	AB3-311	KUMARAN K	NONE	60	9	51
L3+L4	AB1-311	JAYANTHI R	Similar to Registered Slot	60	14	46
L3+L4	AB1-605B	LAKSHMI HARIKA PALIVELA	Registered Slot	60	14	46
L1+L2	AB3-411	GANESH N	NONE	60	16	44
L33+L34	AB1-311	JAYANTHI R	Clashed with D2	60	17	43
L33+L34	AB1-605A	LAKSHMI HARIKA PALIVELA	Clashed with D2	60	18	42
L49+L50	AB1-514	SCOPE6	Clashed with D2	60	18	42
L11+L12	AB1-514	SCOPE13	Same building block in same week day with registered slot L9+L10.	60	27	33
L5+L6	AB1-305B	ALOK CHAUHAN	NONE	60	31	29
L29+L30	AB3-310	BERIN SHALU S	NONE	60	36	24
L11+L12	AB3-311	ALOK CHAUHAN	Different building block in same week day with registered slot L9+L10.	60	60	0

Go Back

Copyright © 2025 Software Development Cell, VIT, Chennai-600 127.

Exhibit D: Property Search

- **Result:** The "Search Property" tab allows any member of the public to query the blockchain for a property's details by its ID, ensuring 100% transparency.

The screenshot shows the Admin Dashboard of the Blockchain Land Registry System. The top navigation bar includes the system name, an ADMIN button, a user ID (0x3d7b9de8584ed23ccbc1ab2abd0d7124be475297), and a Disconnect button. The main navigation menu has four tabs: Register Property, Search Property (selected), Transfer Ownership, and Mortgages & Liens. Below the menu is the Admin Dashboard header. The central section is titled "Search Property Records" and contains two main components: a "Search by Property ID" form and a "Search Information" section. The search form has a text input field for the Property ID (with a placeholder "e.g., 1") and a "Search Property" button. The Search Information section lists the following actions: View detailed property information, Check property verification status, View complete ownership history, Verify document integrity via IPFS hash, and Check for pending transfers.

This screenshot shows the same Admin Dashboard as the previous one, but with the search results for Property ID 8 displayed. The "Search by Property ID" form now has the value "8" in the input field. Below the search form, the "Property Details (ID: 8)" section is visible. It includes a "Status" field with a red "UNVERIFIED" label. The "Current Owner" field displays the address 0x3d7b9De8584Ed23CCBC1ab2aBd0D7124Be475297. The "Location" field shows "Room 919, A Block, VIT Chennai". The "Area" field indicates "1500 sq meters". The "Document Hash (IPFS)" field shows the hash bafybeifnbzu32aenahf4dueh7rfe52bpn530lvge3algao7nsn3loxhfi.

Property Details (ID: 8)

Status:

UNVERIFIED

Current Owner:

0x3d7b9De8584Ed23CCBC1ab2aBd0D7124Be475297

Location:

Room 919, A Block, VIT Chennai

Area:

1500 sq meters

Document Hash (IPFS):

bafybeifnbzu32aenahf4dueh7rfe52bpn53olvge3algao7nsn3ioxfhi

Ownership History

REGISTERED

Owner: 0x3d7b9De8584Ed23CCBC1ab2aBd0D7124Be475297

06/11/2025, 22:40:12

Search Information

Use this section to:

- View detailed property information
- Check property verification status
- View complete ownership history
- Verify document integrity via IPFS hash
- Check for pending transfers

Exhibit E: Admin Verification & System Stats

- **Result:** The "Admin Dashboard" is correctly restricted to the contract owner and provides system-wide statistics ("Total Properties Registered: 8") and the vital "Verify Property" function.

Blockchain Land Registry System

ADMIN0x3d7b9de8584ed23ccbc1ab2abd0d7124be475297Disconnect

Register Property

Search Property

Transfer Ownership

Mortgages & Liens

Admin Dashboard

Admin Dashboard

System Statistics

Total Properties Registered:8

Verify Property

Enter a property ID to approve and validate its registration on the blockchain.

Property ID:

e.g., 1

Verify Property

Admin Responsibilities

- Verify the authenticity of property registrations.
- Ensure all submitted documents are correct and valid.
- Monitor system activity for fraudulent behavior.
- Oversee the integrity of property transfer processes.

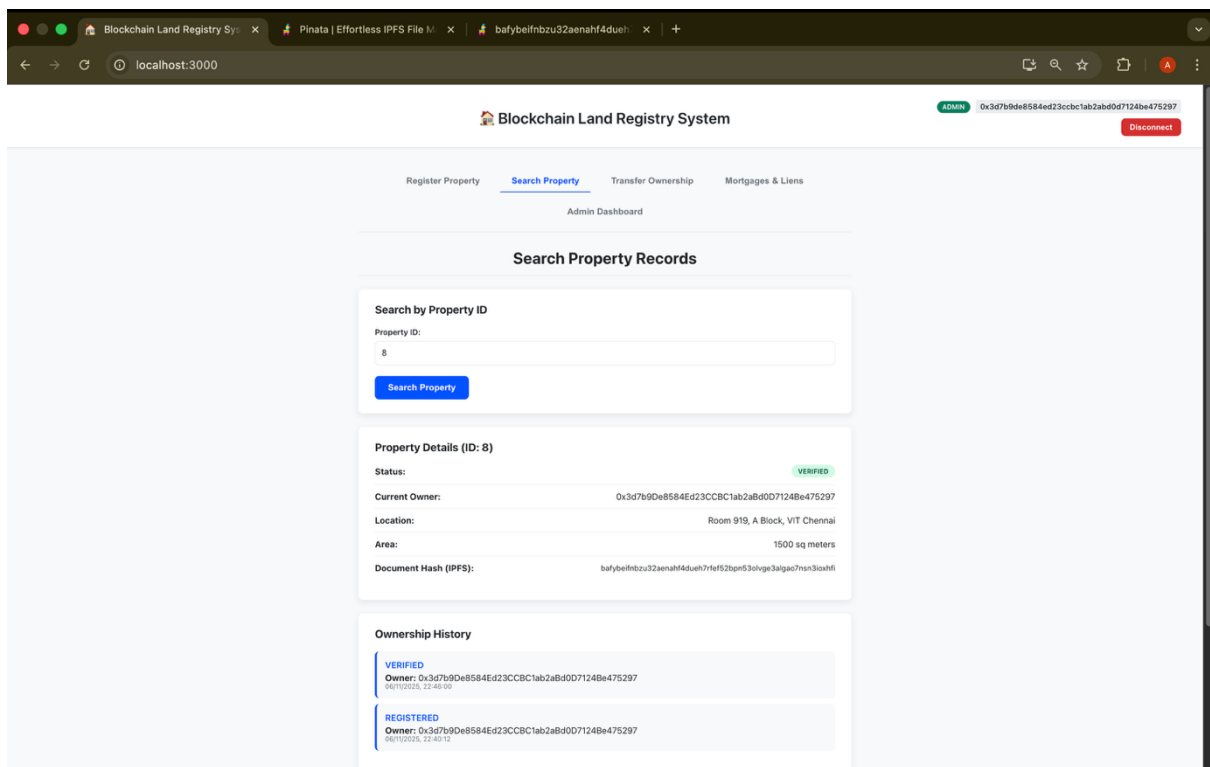
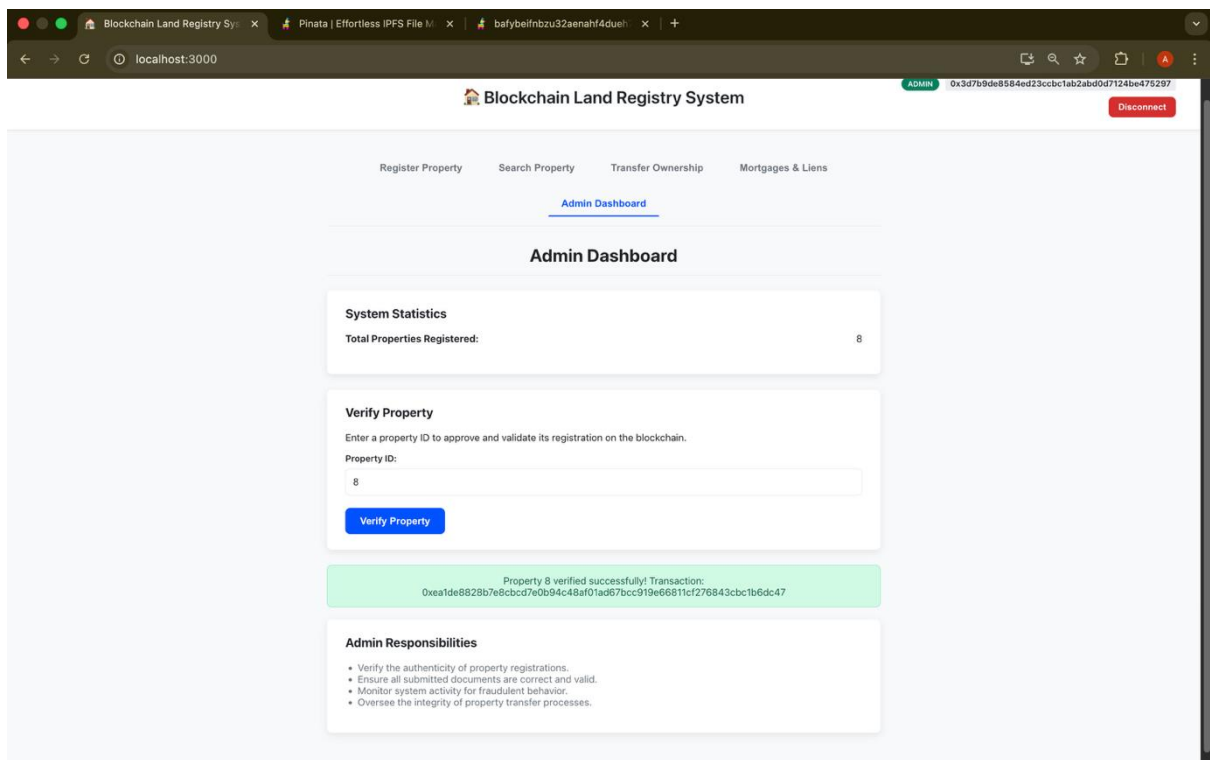


Exhibit E: Ownership Transfer

- **Result:** The "Transfer Ownership" interface clearly separates the three critical actions: Initiate (Seller), Confirm(Buyer), and Cancel (Seller).

The screenshot displays the 'Blockchain Land Registry System' interface. The top navigation bar includes 'Register Property', 'Search Property', 'Transfer Ownership' (highlighted), and 'Mortgages & Liens'. Below this is an 'Admin Dashboard' section. The main content area is titled 'Transfer Property Ownership' and contains three distinct steps:

- Step 1: Initiate Transfer (Seller)**
As the current property owner, you can initiate a transfer to a buyer's address.
Property ID:
Buyer's Wallet Address:
[Initiate Transfer](#)
- Step 2: Confirm Transfer (Buyer)**
As the designated buyer, you must confirm the transfer to complete the process.
Property ID:
[Confirm Transfer](#)
- Cancel Pending Transfer (Seller)**
If you initiated a transfer in error, you can cancel it before the buyer confirms.
Property ID:
[Cancel Transfer](#)

Below these steps is a 'Transfer Process' section with a numbered list:

1. Seller Initiates: Provides property ID and buyer's wallet address.
2. Buyer Confirms: Uses their wallet to confirm the transfer.
3. Ownership Transferred: The change is permanently recorded on the blockchain.
4. Note: Only verified properties can be transferred to ensure security.

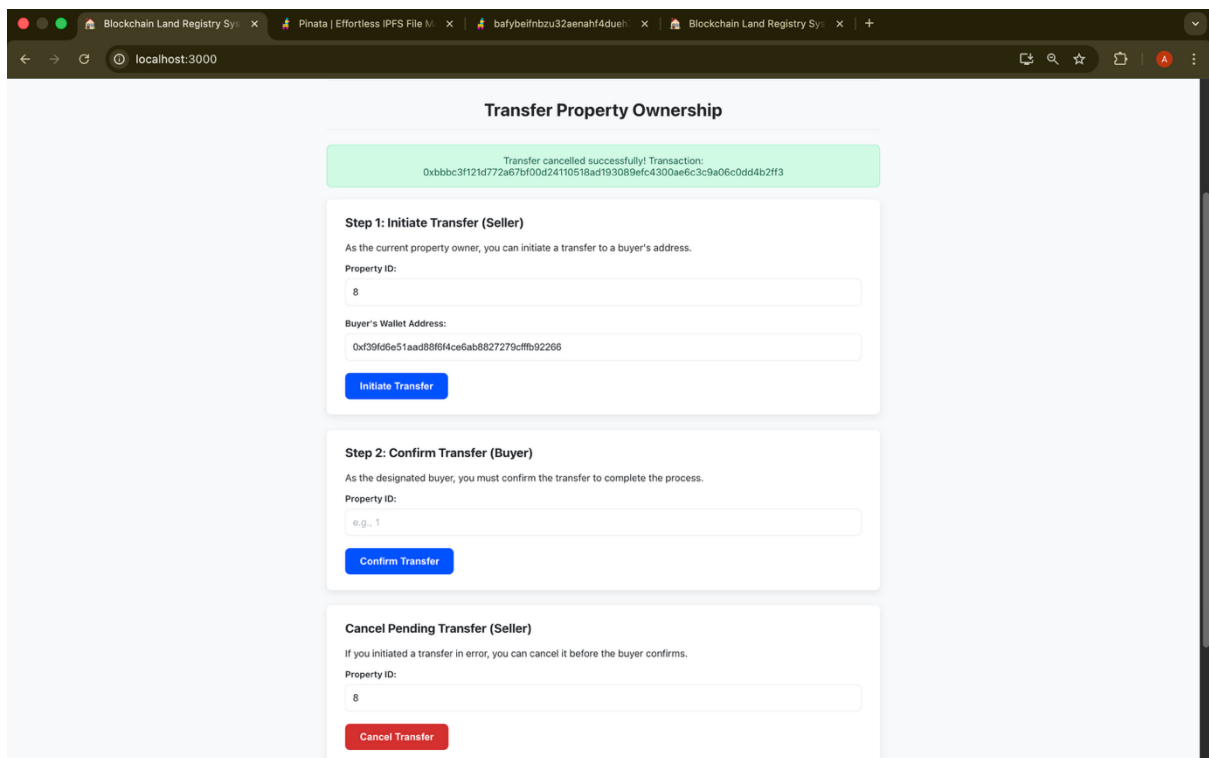
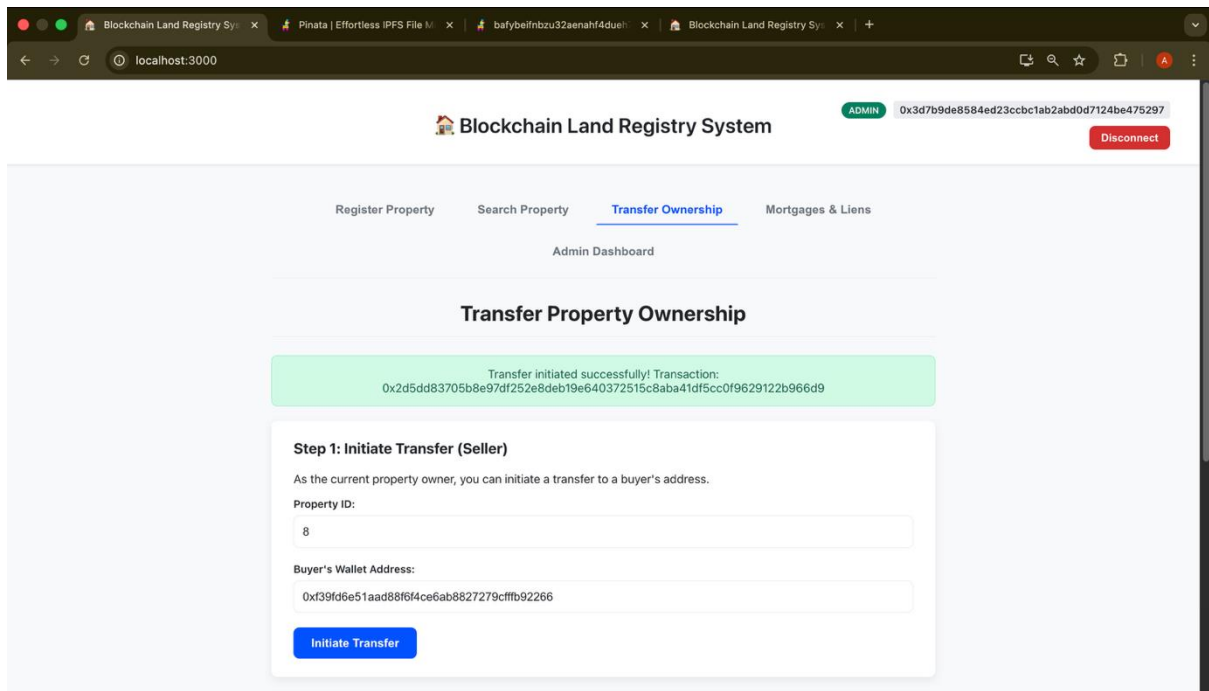


Exhibit G: Mortgage & Lien System

- **Result:** The advanced "Mortgages & Liens" feature is fully implemented, with clear forms for both the *property owner* to place a lien and the *lender* to clear it.

Blockchain Land Registry System

ADMIN 0xf39fd6e51aad88f6f4ce6ab8827279cfff92266 Disconnect

Register Property Search Property Transfer Ownership **Mortgages & Liens**

Admin Dashboard

Manage Mortgages & Liens

Place Lien (Owner)

Property ID
1

Lender Address
0xf39fd6e51aad88f6f4ce6ab8827279cfff92266

Amount (wei)
100

Details / Reference
Loan#123

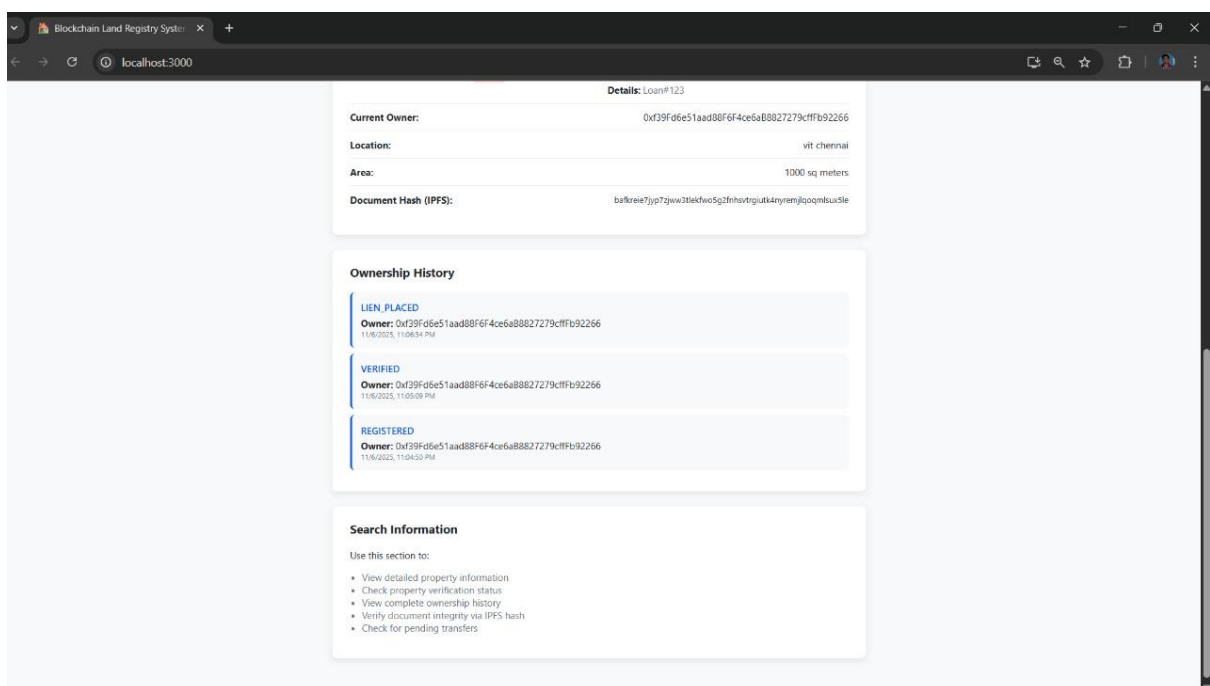
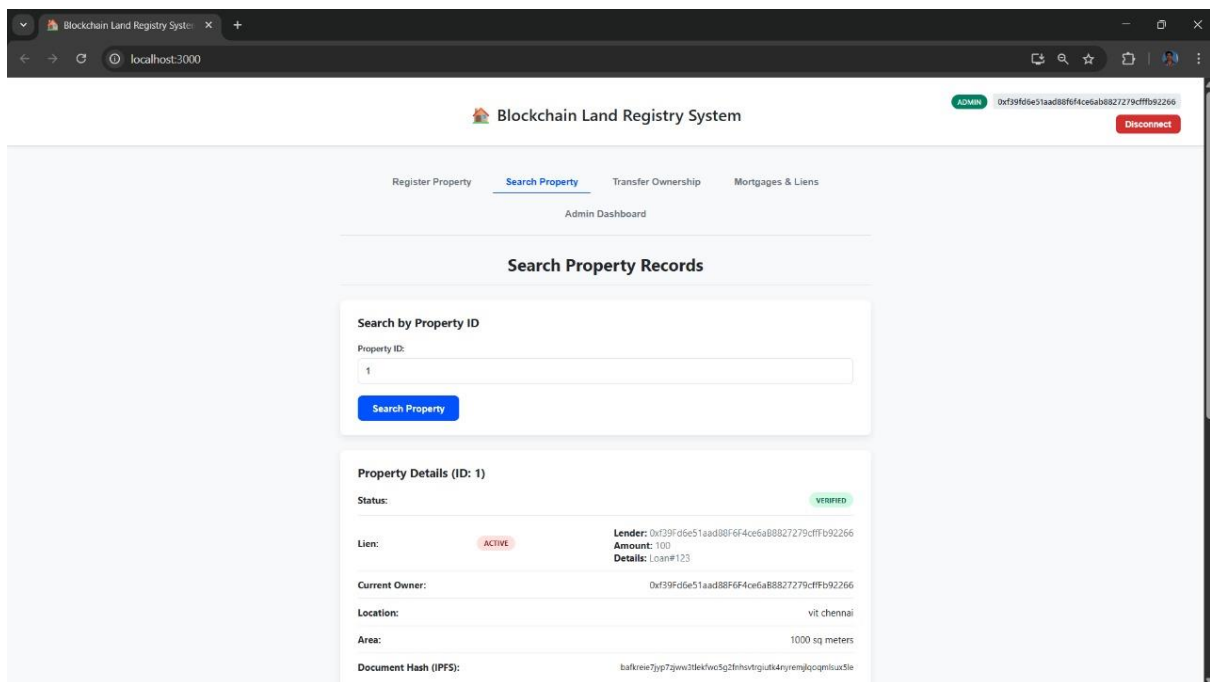
Place Lien

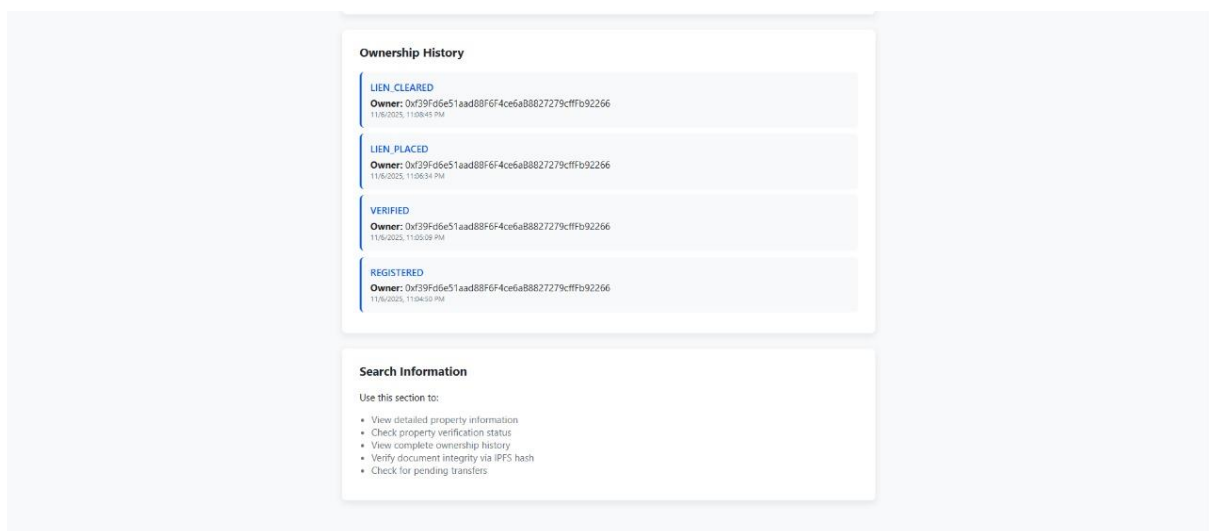
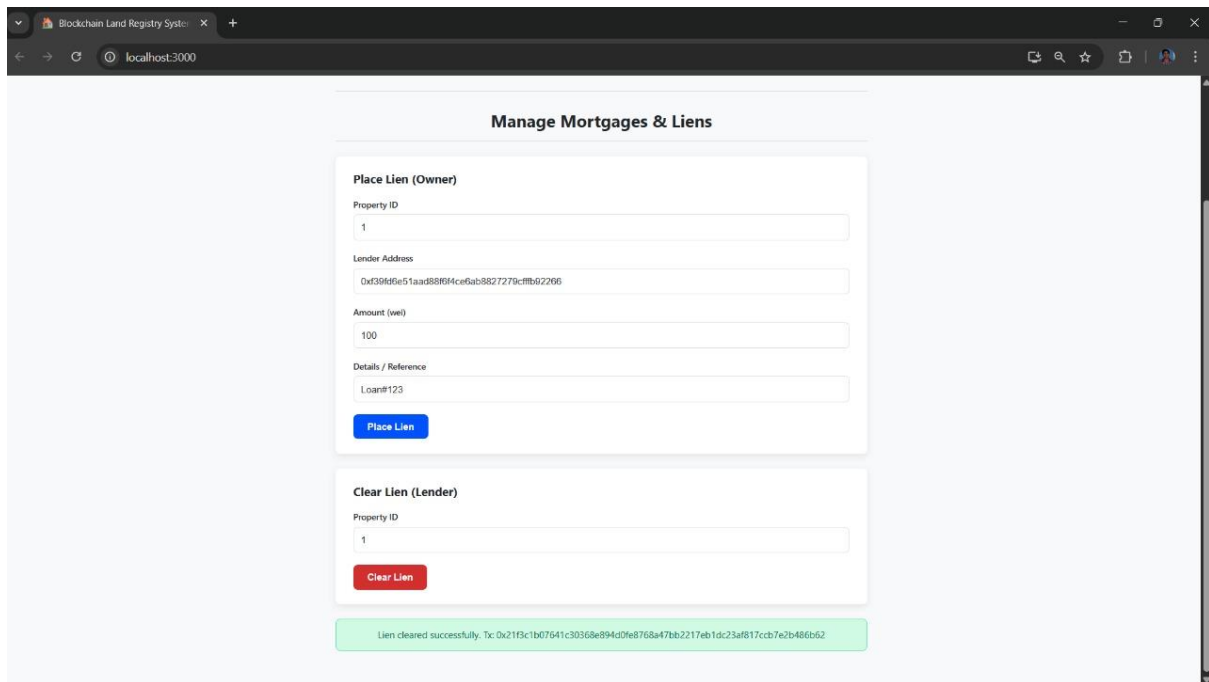
Clear Lien (Lender)

Property ID
1

Clear Lien

Lien placed successfully. Tx: 0x96a0b759449a736dab192d41780edf00d0789f874a16291679dc964da82defe6





Transaction & Test Results

- **Unit Test Success:** All 26 unit tests defined in test/LandRegistry.test.js passed successfully. This includes rigorous testing of all require statements (e.g., onlyOwner, onlyPropertyOwner), edge cases (e.g., transferring to self, clearing a non-existent lien), and the critical security check that blocks the transfer of a property with an active lien.
- **On-Chain Transactions:** All functionalities were successfully executed on the Sepolia testnet, with transactions for registration, verification, transfers, and liens being immutably recorded and verifiable on a block explorer like Etherscan.

Conclusion

This project successfully demonstrates the design and implementation of a robust, secure, and transparent **Blockchain-Based Land Registry System**. By integrating an on-chain Solidity smart contract with decentralized IPFS storage and a modern React frontend, the application solves the core problems of fraud and inefficiency inherent in traditional registry systems.

The successful implementation of all five core features—registration, admin verification, IPFS storage, secure transfer, and the advanced mortgage/lien system—proves that a functional and superior alternative to centralized systems is not only possible but practical. The DApp provides a single source of truth, where all actions are cryptographically signed and publicly auditable, building unprecedented trust into real estate asset management.